

PRISM-Loc: Three LiDAR Localization Backends Behind One ROS 2 Contract, with Middleware-Free Estimator Cores

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Map-based LiDAR localization for mobile robots is solved in fragments: a 2D scanner calls for an AMCL-style particle filter over an occupancy grid, a 3D LiDAR for an NDT-based tracker over a point-cloud map, and an inertial- or GNSS-aided platform for a Kalman fusion node, each with its own topics, parameters, initialization ritual, and recovery behavior. We introduce *PRISM-Loc*, an open-source ROS 2 localization stack that places three established estimator families behind one node contract: `laser2d`, a likelihood-field Monte Carlo localizer; `ndt3d`, an NDT-MCL backend that reuses the same particle filter with a Gaussian-voxel measurement model; and `fusion3d`, a 15-state error-state Kalman filter fusing LiDAR, IMU, and RTK-GNSS. Every backend publishes the same `map→odom` transform and pose with covariance and accepts the same `/initialpose` seeding, all filter mathematics live in Eigen-only C++17 cores that are tested without a ROS runtime, and `laser2d` recovers from the kidnapped-robot condition by deterministic branch-and-bound scan matching over a configurable window around the map center. We claim no new estimator theory; the contribution is architectural. In seeded closed-loop experiments on *synthetic* data that drive the actual cores, the 2D tracker holds a steady-state mean position error of 0.105 m, branch-and-bound relocalization succeeds on 97 of 100 kidnapped-robot queries with zero false accepts at its default threshold, and the ESKF bounds a 10 s aiding dropout to 0.125 m RMSE with a 3σ envelope that the true error never exceeds on the seeded run; no field-accuracy claim is made. Code: https://github.com/kjungmo/prism_loc.

Keywords: mobile robots, LiDAR localization, Monte Carlo localization, NDT, error-state Kalman filter, ROS 2

1. Introduction

Localizing a mobile robot against a prior map is, on paper, a solved problem, but in practice it is solved in fragments. A team fielding a 2D LiDAR reaches for `nav2_amcl`, a particle filter that tracks a planar pose against an occupancy grid and nothing else. A team with a 3D LiDAR reaches instead for an `hdl_localization`-style tracker built around 3D NDT registration (Koide et al., 2019). A team that must fuse inertial and satellite measurements reaches for a `robot_localization`-style node (Moore and Stouch, 2014), which composes an EKF over odometry, IMU, and GNSS but treats the map-relative LiDAR measurement as someone else’s problem. Each of these tools carries its own topic interface, parameter vocabulary, initialization ritual, and recovery story, so a practitioner who moves from a 2D platform to a 3D one, or who adds a GNSS receiver, re-integrates from scratch.

The gap is recognized by the maintainers of the ROS 2 navigation ecosystem. Their survey (Macenski et al., 2023) catalogs the shortcomings of AMCL and sketches, as unbuilt future work, a modular framework of swappable 2D and 3D localization components behind a common interface. The cost

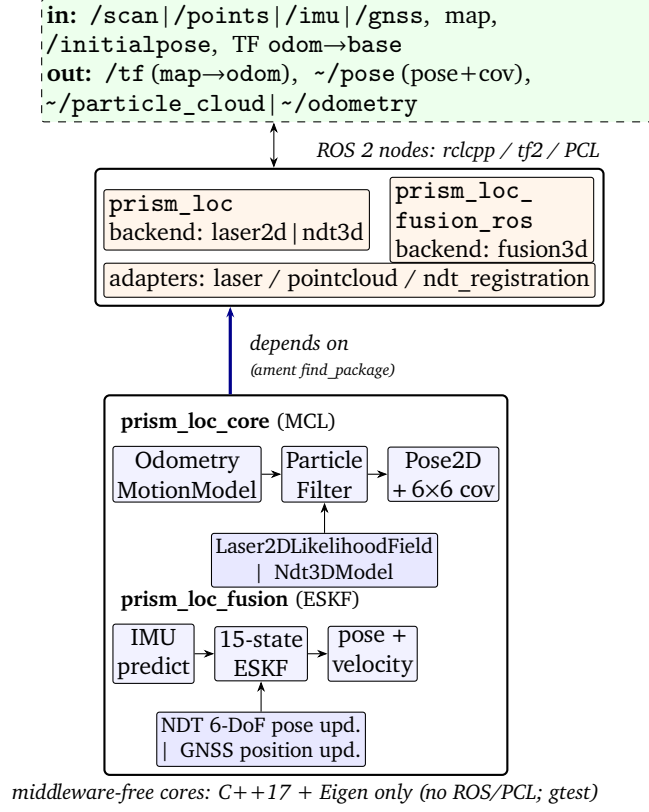


Figure 1: PRISM-Loc overview. Two middleware-free estimator cores (bottom, C++17 + Eigen only): `prism_loc_core` runs the MCL loop (ODOMETRYMOTIONMODEL → PARTICLEFILTER, corrected by either the LASER2DLIKELIHOODFIELD or the NDT3DMODEL measurement model, plus a branch-and-bound relocalizer), and `prism_loc_fusion` runs a 15-state ESKF (IMU prediction, corrected by NDT 6-DoF pose and GNSS position updates). The ROS 2 node packages (top) consume the cores upward through `ament`, select a backend at launch, bridge messages through adapters, and expose one shared contract: a map→odom TF, a pose-with-covariance output, and an /initialpose seed input.

of the gap is not merely inconvenience: the Nav2 marathon deployment (Macenski et al., 2020) attributes a leading share of its autonomy-recovery triggers to loss of localization confidence, so how a stack initializes, degrades, and recovers is a first-class deployment concern rather than a detail. Classical MCL recovers from such failures through stochastic mechanisms, random particle injection or dual sampling (Fox et al., 1999; Thrun et al., 2001), which are hard to reproduce in a test.

To overcome this fragmentation, we introduce *PRISM-Loc*, an engineering response that packages three map-based LiDAR localization backends behind one ROS 2 node contract (Figure 1): `laser2d`, a 2D likelihood-field Monte Carlo localizer against an occupancy grid; `ndt3d`, a 3D NDT-MCL backend that reuses the same particle filter with a voxel-Gaussian measurement model over a prior point cloud, sampling the planar (x, y, θ) pose; and `fusion3d`, a 15-state error-state Kalman filter (ESKF) that fuses 3D LiDAR, IMU, and RTK-GNSS with IMU-bias estimation. All three present the identical interface, a map→odom transform, a pose with 6×6 covariance, and /initialpose seeding, so choosing between a 2D scan, a 3D cloud, and an inertial/GNSS-aided configuration is a launch-time decision rather than a change of stack. We claim no new estimator theory: likelihood-field MCL, NDT-MCL, ESKF fusion, and branch-and-bound scan matching are all established. The contribution is the factoring, and it is delivered as usable open-source software:

- *One Pluggable Contract:* three estimator families (2D likelihood-field MCL, 3D NDT-MCL, and

LiDAR+IMU+RTK-GNSS ESKF fusion) present one ROS 2 map→odom contract, so the sensor suite selects a backend rather than a separate stack; the release adds fail-fast configuration checks, a startup watchdog that names the input that has gone silent, and a single-source parameter reference.

- *Middleware-Free, Deterministic Cores*: all filter mathematics sit behind an Eigen-only boundary with no `roscpp`, `tf2`, or PCL, and all sampling routes through one seeded generator, so every algorithmic claim is exercised by 44 deterministic unit and integration tests and by closed-loop synthetic experiments that compile the cores directly.
- *Deterministic Global Relocalization*: for `laser2d`, branch-and-bound correlative scan matching over a max-pooled likelihood pyramid recovers a kidnapped robot from one scan and re-seeds the MCL posterior, replacing stochastic random-injection recovery with a repeatable search that is provably optimal for its correlative score over the enumerated rotation–translation lattice; the search is global over the map whenever the configured window covers it (by default, maps up to 20 m per axis).

We validate PRISM-Loc at the level of correctness and closed-loop behavior, not field accuracy. On seeded synthetic worlds generated to match the filters’ assumptions, the `laser2d` tracker holds a steady-state mean position error of 0.105 m while KLD-adaptive resampling shrinks the particle set from the 2000-particle ceiling to the 500-particle floor; branch-and-bound relocalization recovers 97 of 100 kidnapped-robot queries under up to 80% beam clutter with zero false accepts at its default acceptance threshold; and the ESKF limits a 10 s loss of all aiding to 0.125 m position RMSE while its own 3σ bound stays above the true error throughout. These numbers demonstrate that the assembled pipelines behave as intended; they are not a benchmark, and quantitative evaluation on public datasets remains future work (Section 5).

2. Related Work

Monte Carlo Localization. Particle-filter localization was introduced by Dellaert et al. (1999) and Fox et al. (1999), who represent the pose posterior as a weighted sample set updated by motion-model prediction, sensor-likelihood weighting, and resampling, and who demonstrated global localization from a uniform prior on real robots. Thrun et al. (2001) later hardened MCL against accurate sensors and the kidnapped-robot problem via Mixture-MCL’s dual sampler, and Fox (2003) made the sample size adaptive through KLD-sampling; the odometry motion model and likelihood-field sensor model we use are the textbook forms of Thrun et al. (2005). The particle-filter core of PRISM-Loc is a direct descendant of this line and of its `nav2_amcl` lineage; we contribute no new filter theory. What differs is factoring: the same predict–weight–resample core is shared, unchanged, between a 2D likelihood-field backend and a 3D NDT backend, and recovery is delegated to a deterministic branch-and-bound search (Section 3.4) rather than stochastic injection or dual sampling.

NDT Representations and Measurement Models. Biber and Straßer (2003) introduced the Normal Distributions Transform, modeling a scan as per-cell Gaussians and reducing registration to Newton optimization of a smooth score; Magnusson et al. (2007) generalized the representation to 3D voxel grids for field deployment on mining vehicles. Both use NDT for deterministic scan-to-scan or scan-to-map registration that requires a reasonable initial guess. The `ndt3d` backend of PRISM-Loc adopts the Gaussian-voxel representation but reads the NDT score out as a per-particle likelihood inside the shared MCL core (NDT-MCL rather than NDT registration), trading the quadratic convergence of Newton alignment for the multi-hypothesis robustness of the particle filter, so the 2D and 3D backends differ only in their measurement model.

LiDAR-Inertial Fusion and Robot Software Systems. Solà (2017) is the mathematical basis of the fusion3d backend: the nominal/error-state decomposition, local angular-error parameterization, and predict–correct–inject–reset cycle are implemented essentially as derived there. On the systems side, Moore and Stouch (2014) established the configurable EKF/UKF fusion node for ROS, Wan et al. (2018) demonstrated an industrial ESKF fusing GNSS-RTK, LiDAR, and IMU at centimeter accuracy as a closed system tied to proprietary maps, and Nav2 (Macenski et al., 2020) treats localization as an external, swappable component. Tightly-coupled LiDAR-inertial odometry such as FAST-LIO (Xu and Zhang, 2021), and applied 3D map-based tracking systems (Koide et al., 2019), are complementary: they produce ego-motion and maps, whereas PRISM-Loc localizes within a prior map. Relative to these, PRISM-Loc places three known estimator families behind one ROS 2 topic and TF contract, each implemented as a library that is unit-testable without a middleware runtime, a property that robot_localization’s otherwise modular design did not prioritize, and which answers the modular-localization framework that Macenski et al. (2023) left as future work.

Global (Kidnapped-Robot) Localization. Olson (2009) formulated 2D scan matching as exhaustive correlative search over a rasterized log-likelihood table with coarse-to-fine multi-resolution pruning, and Hess et al. (2016) bounded that search with branch-and-bound over a precomputed grid pyramid to run loop closure in real time inside Cartographer. PRISM-Loc borrows exactly this branch-and-bound-over-pyramid primitive but applies it in a narrower setting: one-shot relocalization against a known static map, with no submaps or pose graph to maintain, whose result re-seeds the particle filter in place of the heuristic recovery of classical MCL (Fox et al., 1999; Thrun et al., 2001).

3. PRISM-Loc: One Localization Contract over Three Middleware-Free Estimators

PRISM-Loc separates *what* a localization node promises its consumers from *how* a pose is estimated. We first state the estimation problem and the output contract shared by all backends, then describe the two-layer architecture that realizes the contract (Section 3.1), the shared particle-filter core (Section 3.2), its two measurement models (Section 3.3), the branch-and-bound relocalizer (Section 3.4), the error-state fusion filter (Section 3.5), and the initialization and diagnostic layer (Section 3.6). Every equation below is transcribed from the C++ sources listed in Section C.

Problem Setup: One Contract, Interchangeable Estimators. Let m be a prior map (an occupancy grid or a point cloud), $\mathbf{u}_{1:t}$ the odometry or inertial inputs, and $\mathbf{z}_{1:t}$ the exteroceptive measurements (2D scans, 3D clouds, GNSS fixes). Each backend $b \in \mathcal{B} = \{\text{laser2d}, \text{ndt3d}, \text{fusion3d}\}$ approximates the Bayes-filter belief over the robot state $\mathbf{x}_t$ (Thrun et al., 2005),

$$\text{bel}_b(\mathbf{x}_t) = \eta p_b(\mathbf{z}_t | \mathbf{x}_t, m) \int p_b(\mathbf{x}_t | \mathbf{u}_t, \mathbf{x}_{t-1}) \text{bel}_b(\mathbf{x}_{t-1}) d\mathbf{x}_{t-1}, \quad (1)$$

with a particle set for the two MCL backends and a Gaussian error state for fusion3d. Whatever the representation, a backend reports a point estimate $\hat{\mathbf{T}}_{\text{map},\text{base}}$ with a 6×6 covariance $\hat{\Sigma}_t$ and, so as not to fight the odometry source that owns `odom→base_link`, broadcasts only the correction transform

$$\hat{\mathbf{T}}_{\text{map},\text{odom}} = \hat{\mathbf{T}}_{\text{map},\text{base}} \oplus \mathbf{T}_{\text{odom},\text{base}}^{-1}, \quad (2)$$

where $\oplus$ is pose composition and $\mathbf{T}_{\text{odom},\text{base}}$ is read from TF at the measurement stamp. Equation (2) guarantees the round trip $\hat{\mathbf{T}}_{\text{map},\text{odom}} \oplus \mathbf{T}_{\text{odom},\text{base}} = \hat{\mathbf{T}}_{\text{map},\text{base}}$, which ROS-free unit tests assert to 10^{-9} on $SE(2)$ and $SE(3)$. The design objective is that the tuple (TF edge, pose topic, seeding topic) is identical for all $b \in \mathcal{B}$, so that the sensor suite, not the software stack, selects b .

Table 1: Per-backend topic contract. All backends additionally subscribe `/initialpose` and the TF `odom→base_link`, and broadcast `map→odom` by Equation (2) with the stamp future-dated by `transform_tolerance` (default 0.1 s).

Backend	Estimator core	Inputs	Outputs
laser2d	MCL + likelihood field (Equation (7))	<code>/scan</code> , <code>/map</code> (OccupancyGrid)	<code>~/pose</code> , <code>~/particle_cloud</code>
ndt3d	MCL + NDT score (Equation (8))	<code>/points</code> , <code>.pcd</code> map file	<code>~/pose</code> , <code>~/particle_cloud</code>
fusion3d	15-state ESKF (Equations (12) and (13))	<code>/points</code> , <code>/imu</code> , <code>/gnss</code> (NavSatFix), <code>.pcd</code> map file	<code>~/pose</code> , <code>~/odometry</code>

3.1. Separating Estimators from Middleware: Two Layers, One Contract

PRISM-Loc is organized as two layers realized by four packages (Figure 1). The lower layer holds the estimator cores: `prism_loc_core` implements the MCL family and the relocalizer, and `prism_loc_fusion` implements the ESKF. Both are plain C++17 libraries whose only third-party dependency is Eigen; each `CMakeLists.txt` links `Eigen3::Eigen` alone, with no reference to `rclcpp`, `tf2`, or `PCL`. The upper layer holds the ROS 2 node packages `prism_loc` (the `laser2d` and `ndt3d` backends, selected by a backend string parameter) and `prism_loc_fusion_ros` (the `fusion3d` backend). Only these depend on the middleware, and the dependency direction is strictly upward.

The split buys three properties. First, *unit-testability without a running graph*: each core builds and tests with a plain toolchain and GoogleTest, and continuous integration exercises this ROS-free path directly. Second, *determinism*: all sampling routes through a single seeded `std::mt19937_64` wrapper (ROS parameter `seed`, default 42), so a prediction, weighting, or resampling step is reproducible bit for bit, which makes the core tests assertable rather than merely statistical. Third, *one output contract* (Table 1): every backend broadcasts `map→odom` per Equation (2), publishes a `PoseWithCovarianceStamped` on `~/pose`, and accepts `/initialpose`; the MCL nodes add a particle cloud and the fusion node an `Odometry` message. Thin adapters convert `LaserScan`, `OccupancyGrid`, `PointCloud2`, and `.pcd` files into core value types, and in `fusion3d` an adapter wraps PCL’s NDT registration to turn a live cloud into a 6-DoF pose measurement, so PCL appears only inside adapters. QoS profiles follow publisher semantics: sensor streams are subscribed with `SensorDataQoS` (best-effort), while the occupancy grid is subscribed `transient_local` and `reliable` so the single latched map from `map_server` reaches a late-joining node.

3.2. Sampling and Weighting Particles: One MCL Core for 2D and 3D

Both pose-tracking backends share one particle filter over the planar pose $\mathbf{x} = (x, y, \theta)$ (Pose2D), with belief $\{(\mathbf{x}_t^{[i]}, w_t^{[i]})\}_{i=1}^{N_t}$. The recursion is the one wired in `localization_node.cpp` and summarized in Algorithm 1.

Motion Model. `OdometryMotionModel` implements the sampling form of the odometry model (Thrun et al., 2005). Consecutive odometry poses are decomposed into $(\delta_{r1}, \delta_t, \delta_{r2})$, with $\delta_{r1} = 0$ whenever $\delta_t < 10^{-3}$ m (pure-rotation guard), and each particle is advanced by the noise-corrupted increment

$$\begin{aligned}\hat{\delta}_{r1} &= \delta_{r1} - \varepsilon(\alpha_1 \delta_{r1}^2 + \alpha_2 \delta_t^2), & \hat{\delta}_{r2} &= \delta_{r2} - \varepsilon(\alpha_1 \delta_{r2}^2 + \alpha_2 \delta_t^2), \\ \hat{\delta}_t &= \delta_t - \varepsilon(\alpha_3 \delta_t^2 + \alpha_4 (\delta_{r1}^2 + \delta_{r2}^2)),\end{aligned}\tag{3}$$

Algorithm 1: One laser2d/ndt3d filter update, as wired in `localization_node.cpp`

Input: particle set $\mathcal{X}$; previous and current odometry poses $\mathbf{o}_{t-1}, \mathbf{o}_t$; measurement model ℓ ;
gates $d_{\min}, a_{\min}$

```

1 if  $\|\mathbf{o}_t - \mathbf{o}_{t-1}\|_{xy} \geq d_{\min}$  or  $|\theta(\mathbf{o}_t) - \theta(\mathbf{o}_{t-1})| \geq a_{\min}$  or a re-seed forced an update then
2   for  $i \leftarrow 1$  to  $|\mathcal{X}|$  do
3      $\mathbf{x}^{[i]} \leftarrow$  sample Equation (3) from  $(\mathbf{o}_{t-1}, \mathbf{o}_t)$  applied to  $\mathbf{x}^{[i]}$ ;
4     compute weights  $w^{[i]}$  by Equation (4);
5     if  $N_{\text{eff}} < \tau|\mathcal{X}|$  then
6        $\mathcal{X} \leftarrow$  KLD-resample until  $n \geq \max(M_k, N_{\min})$  (Equation (5)) or  $n = N_{\max}$ ;
7      $\mathbf{o}_{t-1} \leftarrow \mathbf{o}_t$ ;

```

Output: $(\hat{x}, \hat{y}, \hat{\theta})$ and covariance by Equation (6); $\hat{\mathbf{T}}_{\text{map,odom}}$ by Equation (2)

where $\varepsilon(\sigma^2)$ draws zero-mean Gaussian noise of variance σ^2 and $\alpha_{1..4}$ are the alpha1–alpha4 parameters.

Weighting. Given a measurement model with log-likelihood $\ell(\mathbf{x})$ (Section 3.3), the correct step of `ParticleFilter` replaces each weight by

$$w^{[i]} = \frac{\exp(\ell(\mathbf{x}^{[i]}) - \ell_{\max})}{\sum_j \exp(\ell(\mathbf{x}^{[j]}) - \ell_{\max})}, \quad \ell_{\max} = \max_j \ell(\mathbf{x}^{[j]}), \quad (4)$$

subtracting the per-step maximum for numerical stability. The weights are overwritten rather than multiplied into the previous ones; this coincides with the standard recursion after every resampling step (which leaves uniform weights), and otherwise discards the previous step’s weights, a simplification we note rather than hide.

KLD-Adaptive Resampling. Resampling is triggered only when the effective sample size $N_{\text{eff}} = (\sum_i w^{[i]})^2 / \sum_i (w^{[i]})^2$ falls below τN (`resample_threshold` $\tau = 0.5$). Samples are then drawn with probability proportional to weight and dropped into a histogram with bins of `kld_bin_xy` in (x, y) and `kld_bin_yaw` in θ . Following Fox (2003), after $k > 1$ distinct bins have been occupied the target sample count is

$$M_k = \frac{k-1}{2\epsilon_{\text{kld}}} \left(1 - \frac{2}{9(k-1)} + \sqrt{\frac{2}{9(k-1)}} z_{1-\delta} \right)^3, \quad (5)$$

and drawing stops once $n \geq \max(M_k, N_{\min})$ or $n = N_{\max}$, with ϵ_{kld} (`kld_err`), $z_{1-\delta}$ (`kld_z`), and $[N_{\min}, N_{\max}]$ (`min_particles`, `max_particles`) as parameters. The resampled set receives uniform weights.

Pose Estimate. The reported pose is the weighted mean of (x, y) with the circular mean of θ ,

$$\hat{x} = \sum_i w^{[i]} x^{[i]}, \quad \hat{y} = \sum_i w^{[i]} y^{[i]}, \quad \hat{\theta} = \text{atan2}\left(\sum_i w^{[i]} \sin \theta^{[i]}, \sum_i w^{[i]} \cos \theta^{[i]}\right), \quad (6)$$

and the 6×6 covariance carries the weighted planar second moments in (x, y) and θ , while z , roll, and pitch are marked unobserved with variance 10^6 .

3.3. Scoring Observations: Likelihood-Field and NDT Measurement Models

A backend differs from the other only through a `MeasurementModel`, whose single virtual method returns the unnormalized log-likelihood $\ell(\mathbf{x})$ of the current observation given a candidate base pose.

2D Likelihood Field. Laser2DLikelihoodField is the likelihood-field model (Thrun et al., 2005). At construction the occupancy grid (cells with value ≥ 50 occupied) is converted into a field $d(\cdot)$ holding the Euclidean distance to the nearest occupied cell, computed exactly by the separable squared-distance transform of Felzenszwalb and Huttenlocher (2012) and clamped at `likelihood_max_dist`. For a scan with ranges r_j at bearings ϕ_j , the model visits every s -th beam, $s = \max(1, \lfloor n/\text{max_beams} \rfloor)$, discards ranges outside $(r_{\min}, r_{\max})$, and scores

$$\ell_{2D}(\mathbf{x}) = \sum_{j \in \mathcal{J}} \log \max \left(z_{\text{hit}} \exp(-d(\mathbf{e}_j(\mathbf{x}))^2 / 2\sigma_{\text{hit}}^2) + z_{\text{rand}}/r_{\max}, 10^{-12} \right), \quad (7)$$

where $\mathbf{e}_j(\mathbf{x})$ is the beam endpoint $r_j(\cos \phi_j, \sin \phi_j)$ transformed into the map through $\mathbf{x}$ composed with the sensor mounting pose.

3D NDT Score. Ndt3DModel scores a 3D point cloud against a Normal Distributions Transform map (Biber and Straßer, 2003; Magnusson et al., 2007). NdtMap voxelizes the prior cloud at resolution `ndt_resolution`; each voxel v with at least `voxel_min_points` points stores its mean μ_v and the inverse of its covariance after inflating every eigenvalue to at least 10^{-3} of the largest. With $\mathbf{T}(\mathbf{x})$ the 3D lift of the planar particle at the fixed height `base_height` composed with the sensor pose, the per-particle score over a subsampled cloud $\mathcal{P}$ (at most `max_points` points) is

$$\ell_{3D}(\mathbf{x}) = \sum_{\mathbf{p} \in \mathcal{P}} \log(s(\mathbf{T}(\mathbf{x})\mathbf{p}) + s_{\text{floor}}), \quad s(\mathbf{q}) = \exp\left(-\frac{1}{2}(\mathbf{q} - \mu_{v(\mathbf{q})})^\top \Sigma_{v(\mathbf{q})}^{-1}(\mathbf{q} - \mu_{v(\mathbf{q})})\right), \quad (8)$$

with $s(\mathbf{q}) = 0$ when $\mathbf{q}$ falls in an empty voxel and $s_{\text{floor}} = 10^{-6}$. This is the NDT-as-measurement-model (NDT-MCL) idea: the score that Newton registration usually optimizes is read out as a particle likelihood, and the 3D backend estimates the same (x, y, θ) as the 2D one.

3.4. Relocalizing by Branch-and-Bound: Deterministic Kidnapped-Robot Recovery

Recovery from a lost or kidnapped estimate is handled by BranchAndBoundMatcher, a correlative scan matcher (Olson, 2009) bounded by a max-pooled pyramid as in Hess et al. (2016). At construction the likelihood field (clamped at $4\sigma_{\text{hit}}$) is converted into a probability grid and a pyramid of $D + 1$ levels (`bbs_max_depth` D),

$$P_0(c) = \exp(-d(c)^2 / 2\sigma_{\text{hit}}^2), \quad P_h(c_x, c_y) = \max_{0 \leq a, b < 2^h} P_0(c_x + a, c_y + b), \quad (9)$$

where cells outside the grid count as zero; P_h is built by doubling, $P_h(c) = \max\{P_{h-1}(c), P_{h-1}(c + 2^{h-1}\mathbf{e}_x), P_{h-1}(c + 2^{h-1}\mathbf{e}_y), P_{h-1}(c + 2^{h-1}\mathbf{1})\}$. A search at center $\mathbf{c}$ (the map center in the node) enumerates rotations θ_k over $\pm \text{bbs_angular_window}$ at `bbs\_angular\_step` and integer cell offsets $\mathbf{o}$ starting at $-L$, $L = \text{round}(W/\text{res})$ for linear window W . Because the root nodes are anchored at $-L, -L + 2^D, \dots \leq L$ and each covers 2^D offsets per axis, the offsets actually enumerated form $O = [-L, -L + 2^D \lceil (2L + 1)/2^D \rceil]^2$, a superset of $[-L, L]^2$ that is asymmetric (for the defaults, $W = 10$ m at 0.1 m and $D = 6$, it spans -10 m to $+15.5$ m about $\mathbf{c}$). With Θ the rotation set, the search space is $\Theta \times O$. For rotation θ_k , let $\mathbf{g}_j^k$ be the grid cell of the j -th subsampled endpoint ($j \in \mathcal{E}$, at most `bbs_max_beams`) at the center translation. A node at depth h with anchor $\mathbf{o}$ covers the offsets $\mathbf{o} + [0, 2^h]^2$ and is scored by

$$S_h(k, \mathbf{o}) = \sum_{j \in \mathcal{E}} P_h(\kappa_h(\mathbf{g}_j^k + \mathbf{o})), \quad (10)$$

where at $h = 0$ out-of-grid cells contribute zero, and at $h \geq 1$ a block that lies entirely outside the grid contributes zero while any other anchor is clamped into the grid by κ_h . Leaves ($h = 0$) carry the exact correlative score S_0 , which is the correlative surrogate of Hess et al. (2016); Olson (2009), not the MCL log-likelihood of Equation (7).

Algorithm 2: Branch-and-bound global relocalization (bbs . cpp)

Input: scan endpoints $\mathcal{E}$; pyramid $\{P_h\}_{h=0}^D$; rotations $\{\theta_k\}$; window L ; acceptance fraction ρ

```

1  $C \leftarrow \{(k, \mathbf{o}, D, S_D(k, \mathbf{o})) : \forall k, \mathbf{o} \in \{-L, -L + 2^D, \dots\}^2, \mathbf{o} \leq L\};$ 
2  $S^* \leftarrow -1;$ 
3 Procedure Branch( $C$ )
4   sort  $C$  by bound, descending;
5   foreach  $(k, \mathbf{o}, h, U) \in C$  do
6     if  $U \leq S^*$  then break // admissible prune;
7     if  $h = 0$  then
8        $S^* \leftarrow U; \mathbf{x}^* \leftarrow (\mathbf{c} + \text{res} \cdot \mathbf{o}, \theta_k);$ 
9     else
10      Branch( $\{(k, \mathbf{o} + \boldsymbol{\delta}, h - 1, S_{h-1}(k, \mathbf{o} + \boldsymbol{\delta})) : \boldsymbol{\delta} \in \{0, 2^{h-1}\}^2\}$ );
11 Branch( $C$ );
Output:  $\mathbf{x}^*$ , valid iff  $S^* \geq \rho|\mathcal{E}|$  (Equation (11))
```

Proposition 1 (Admissible bound). *For every node $(k, \mathbf{o})$ at depth $h \geq 1$ and every leaf offset $\mathbf{o}' \in \mathbf{o} + [0, 2^h)^2$, $S_h(k, \mathbf{o}) \geq S_0(k, \mathbf{o}')$.*

Proof. Fix an endpoint j and let $\mathbf{c} = \mathbf{g}_j^k + \mathbf{o}$. If the block $\mathbf{c} + [0, 2^h)^2$ lies entirely outside the grid, every leaf cell $\mathbf{g}_j^k + \mathbf{o}'$ is also outside and contributes zero, so both sides are 0. Otherwise, when $\mathbf{c}$ is inside the grid $P_h(\mathbf{c})$ is by Equation (9) the maximum of P_0 over the block, which contains the leaf cell. When a coordinate of $\mathbf{c}$ is negative, clamping it to 0 yields a block $[0, 2^h)$ in that coordinate that contains every in-grid cell of $[c, c + 2^h)$, because $c < 0$; out-of-grid leaf cells contribute $0 \leq P_h$. Hence each endpoint's term at depth h upper-bounds its term at every leaf of the node, and summing over j gives the claim. (Without the clamp, a coarse node whose corner falls off the map would under-count and could prune the optimum; the implementation comment in bbs . cpp records exactly this case.) $\square$

Algorithm 2 expands nodes depth-first with children visited in decreasing order of their bound and prunes a subtree whenever its bound does not exceed the best leaf score found so far. By Proposition 1 no pruned subtree can contain a strictly better leaf, so the returned pose attains the maximum of S_0 over $\Theta \times \mathcal{O}$ (ties go to the first leaf reached). The guarantee holds for pyramid values stored in single precision and summed in double: each per-endpoint term at depth h is an exact maximum of leaf values, and the endpoints are summed in the same order at every depth, so monotonicity of rounded addition preserves the inequality. It is optimality on this lattice only, not over continuous poses. It is *global* only when $\mathbf{c} + \text{res} \cdot \mathcal{O}$ covers the map: the node fixes $\mathbf{c}$ at the map center and does not size W to the map, so with the default W a robot farther than 10 m from the center on the negative side of an axis is outside the search. The match is accepted only when

$$S_0^* \geq \rho|\mathcal{E}|, \quad (11)$$

with $\rho = \text{bbs_min_score_fraction}$ and $|\mathcal{E}|$ the number of valid endpoints used; an accepted pose re-seeds the particle filter with a Gaussian of standard deviation (0.2 m, 0.2 m, 0.1 rad), whereas a rejected one leaves the estimate untouched. The search is deterministic and therefore reproducible in a unit test.

3.5. Fusing IMU, LiDAR, and GNSS: A 15-State Error-State Kalman Filter

The `fusion3d` backend runs `Eskf`, which follows the quaternion error-state formulation of Solà (2017). The nominal state carries position $\mathbf{p}$, velocity $\mathbf{v}$, orientation $\mathbf{q}$, and accelerometer and gyroscope biases $\mathbf{b}_a, \mathbf{b}_g$; the error state is $\delta\mathbf{x} = (\delta\mathbf{p}, \delta\mathbf{v}, \delta\theta, \delta\mathbf{b}_a, \delta\mathbf{b}_g) \in \mathbb{R}^{15}$ with a local angular error $\delta\theta$.

Prediction. On each IMU sample $(\mathbf{a}_m, \omega_m)$ with period Δt , let $\mathbf{R} = \mathbf{R}(\mathbf{q})$, $\mathbf{a} = \mathbf{a}_m - \mathbf{b}_a$, $\omega = \omega_m - \mathbf{b}_g$, and $\mathbf{a}_w = \mathbf{R}\mathbf{a} + \mathbf{g}$ with $\mathbf{g} = (0, 0, -9.81)$. The nominal state and error covariance evolve as

$$\mathbf{p} += \mathbf{v}\Delta t + \frac{1}{2}\mathbf{a}_w\Delta t^2, \quad \mathbf{v} += \mathbf{a}_w\Delta t, \quad \mathbf{q} \leftarrow \mathbf{q} \otimes \text{Exp}(\omega\Delta t), \quad \mathbf{P} \leftarrow \mathbf{F}\mathbf{P}\mathbf{F}^\top + \mathbf{Q}, \quad (12)$$

where Exp is the $SO(3)$ exponential, $\mathbf{F}$ is the 15×15 identity except for the blocks

$$\mathbf{F}_{pv} = \mathbf{I}\Delta t, \quad \mathbf{F}_{v\theta} = -\mathbf{R}[\mathbf{a}]_\times\Delta t, \quad \mathbf{F}_{vb_a} = -\mathbf{R}\Delta t, \quad \mathbf{F}_{\theta\theta} = \mathbf{R}\{\text{Exp}(\omega\Delta t)\}^\top, \quad \mathbf{F}_{\theta b_g} = -\mathbf{I}\Delta t,$$

and the process noise

$$\mathbf{Q} = \text{blkdiag}(\mathbf{0}_3, \sigma_a^2\Delta t^2\mathbf{I}_3, \sigma_g^2\Delta t^2\mathbf{I}_3, \sigma_{b_a}^2\Delta t\mathbf{I}_3, \sigma_{b_g}^2\Delta t\mathbf{I}_3)$$

is built from `sigma_acc`, `sigma_gyro`, `sigma_acc_bias`, and `sigma_gyro_bias`; $\mathbf{R}\{\cdot\}$ denotes the rotation matrix of a quaternion.

Correction, Injection, Reset. Two updates share the Kalman gain,

$$\mathbf{K} = \mathbf{P}\mathbf{H}^\top(\mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R}_z)^{-1}, \quad \delta\hat{\mathbf{x}} = \mathbf{K}\mathbf{y}, \quad \mathbf{P} \leftarrow (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}. \quad (13)$$

`updatePose` applies a 6-DoF pose from NDT registration with $\mathbf{H}$ selecting $(\delta\mathbf{p}, \delta\theta)$ and residual $\mathbf{y} = (\mathbf{p}_m - \mathbf{p}, \text{Log}(\mathbf{q}^{-1} \otimes \mathbf{q}_m))$, with $\mathbf{R}_z = \text{diag}(\sigma_{\text{pos}}^2\mathbf{I}, \sigma_{\text{rot}}^2\mathbf{I})$ (`pose_pos_std`, `pose_rot_std`); the node applies it only when PCL's NDT converged with fitness at most `ndt_max_fitness`. `updatePosition` applies a 3-DoF GNSS fix, converted from WGS-84 through ECEF to a local ENU frame about a datum (the first accepted fix unless `use_datum` fixes one), with $\mathbf{H}$ selecting $\delta\mathbf{p}$ and $\mathbf{R}_z$ taken from the fix covariance (horizontal variances floored at 10^{-4} , vertical at 1). After each update the error is injected, additively for $\mathbf{p}, \mathbf{v}, \mathbf{b}_a, \mathbf{b}_g$ and as $\mathbf{q} \leftarrow \mathbf{q} \otimes \text{Exp}(\delta\hat{\theta})$ for orientation, and the error mean is reset to zero. The implementation uses the simple covariance update of Equation (13) and omits the reset Jacobian (it takes $\mathbf{G} = \mathbf{I}$), a common first-order simplification of the full formulation in Solà (2017).

3.6. Initializing and Diagnosing: Making Every Precondition Observable

All backends accept a `PoseWithCovarianceStamped` on `/initialpose`: the MCL backends spread a Gaussian particle cloud at the supplied mean and covariance, and the fusion backend loads it as its nominal state. Automatically, `laser2d` runs Algorithm 2 before its first pose when `try_global_localization` is set, or on demand through the `~/global_localization` service (the matcher is constructed lazily at the first call). `fusion3d` auto-initializes once it holds both an attitude, roll and pitch from the first IMU specific force $\mathbf{f}$ and heading ψ_0 from `initial_yaw`,

$$\phi_0 = \text{atan2}(f_y, f_z), \quad \vartheta_0 = \text{atan2}(-f_x, \sqrt{f_y^2 + f_z^2}), \quad \mathbf{q}_0 = \mathbf{R}_z(\psi_0) \mathbf{R}_y(\vartheta_0) \mathbf{R}_x(\phi_0), \quad (14)$$

and a position from the first GNSS fix that passes the gate $\{\text{status} \geq s_{\min}\} \wedge \{\text{lat, lon not NaN}\} \wedge \{c_{xx} \leq c_{\max}\}$ (`gnss_min_status`, `gnss_max_pos_cov`).

Because a localization node that silently does nothing is the dominant first-run frustration, four diagnostic mechanisms accompany the estimators: (i) `ndt3d` and `fusion3d` abort construction with

an actionable error when `map_pcd_path` is empty or fails to load; (ii) every callback that cannot yet act emits a rate-limited warning naming the unmet precondition and the remedy; (iii) a 10 s startup watchdog reports each silent mandatory input together with its live publisher count, separating “no publisher” from “publisher present but QoS-incompatible”; and (iv) the GNSS gate and the NDT update log each rejection with the measured value against the violated threshold. Every tunable is a declared ROS 2 parameter documented one-to-one in `PARAMS.md` (Section B). These mechanisms carry no algorithmic weight; they exist so that a misconfiguration is diagnosable from the console log alone, consistent with the operational transparency Nav2 found necessary for long-term autonomy (Macenski et al., 2020).

4. Experiments

We validate PRISM-Loc at the level of *correctness and closed-loop behavior*, not field accuracy. Section 4.1 describes the test suite and the three synthetic closed-loop experiments, Section 4.2 reports their results, and Section 4.3 isolates the contribution of individual mechanisms using the same committed runs. All data in this section are synthetic, and we make no accuracy claim about physical robots.

4.1. Experimental Setup

Correctness Tests. The suite comprises 44 GoogleTest cases across the four packages: 25 in `prism_loc_core` (both MCL backends and global recovery), 13 in `prism_loc_fusion` (the ESKF), 4 in the `prism_loc` adapter layer, and 2 in `prism_loc_fusion_ros`. The motion model is checked deterministically (zero-command and pure-translation cases) and statistically (the mean of noisy samples approximates the commanded motion); both measurement models are checked for the monotonicity a likelihood must satisfy (the true pose out-scores a displaced one); resampling, the weighted-mean estimate, the grid and NDT-map primitives, and branch-and-bound recovery of a single-scan kidnapped pose (with rejection of an empty scan) are tested directly; two integration tests close the `laser2d` and `ndt3d` loops until convergence to ground truth. The ESKF is pinned for rest under a static IMU, constant-acceleration kinematics, measurement pull-in, and tracking of a moving target from a wrong initial guess; $SO(3)$ exponential/logarithm round trips, geodetic ENU round trips, and the Equation (2) composition are verified separately, and RNG tests assert that equal seeds reproduce identical sequences. Two CI jobs guard the boundary from both sides: a *core* job builds the two cores with plain CMake and runs `ctest` with only a compiler and Eigen, and a *ROS* job runs `colcon build` and `colcon test` for all four packages in a `ros:humble` container.

Synthetic Closed-Loop Protocol. Three drivers under `experiments/` compile the *actual* core sources directly with `g++` and Eigen (no ROS, no CMake) and generate their own worlds, scans, and inertial streams, with all randomness derived from fixed compile-time seeds. The sensor-noise models match the filters’ assumptions by construction, so the experiments demonstrate behavior rather than benchmark accuracy.

- *MCL tracking.* A $12\text{ m} \times 10\text{ m}$ grid at 0.1 m/cell (room border plus two interior pillars) hosts a robot driven around a closed oval loop with semi-axes 3.6 m and 2.6 m for 65 s at 10 Hz (650 steps). Each step ray-casts a 180-beam scan at the true pose and synthesizes drifting odometry from noise-corrupted increments under the `laser2d.yaml` model ($\alpha_{1..4} = 0.2$). The filter, with every parameter from `laser2d.yaml`, is seeded with a broad Gaussian prior ($\sigma = 1\text{ m}$, 0.4 rad) at the true start and updated as in Algorithm 1 with $d_{\min} = a_{\min} = 0.2$. Seeds: 101, 202, 303.
- *BBS relocation.* On an asymmetric $6\text{ m} \times 6\text{ m}$ room (two interior wall stubs break the four-fold

Table 2: laser2d steady-state tracking error on synthetic data ($t \geq 10$ s, 110 logged samples per seed at 2 Hz). Columns are computed from `data/mcl_tracking.csv`; the last two rows pool the three seeds as mean $\pm$ std of the per-seed values and as statistics over all 330 samples.

Seed	Mean pos. err. (m)	Max pos. err. (m)	Mean yaw err. (rad)	Max yaw err. (rad)
101	0.113	0.269	0.034	0.120
202	0.100	0.257	0.030	0.105
303	0.102	0.288	0.033	0.184
Per-seed mean $\pm$ std	0.105 $\pm$ 0.007	–	0.032 $\pm$ 0.002	–
Pooled (330 samples)	0.105	0.29	0.032	0.18

symmetry, the world of the BBS unit test), $N = 100$ poses are drawn uniformly over free space with uniform yaw, and one 180-beam scan is ray-cast per pose. To span easy to hard cases, each beam is replaced with a uniform-random spurious range with a per-query probability drawn uniformly in $[0, 0.8]$, modeling people, glass, and objects absent from the map. The real matcher runs with the `bbs_*` defaults ($W = 10$ m, $\pm\pi$ at 0.0175 rad $\approx 1^\circ$, $D = 6$, 120 beams), centered on the room, so its window covers the whole map; this study therefore does not exercise maps larger than the window (Section 5). Seed: 20240607.

- *ESKF under aiding dropout.* The real ESKF runs over a 60 s 3D trajectory, a horizontal figure-eight $x = 8 \sin \omega t$, $y = 4 \sin 2\omega t$ with $\omega = 2\pi/30$ and a ± 1.5 m altitude oscillation. The analytic specific force and angular rate are corrupted with exactly the white-noise densities and constant biases the process model assumes (`fusion3d.yaml` defaults), plus fixed true biases the filter must estimate from zero. The IMU runs at 200 Hz, a simulated NDT source delivers 6-DoF poses at 10 Hz and a simulated GNSS source positions at 1 Hz, with noise matched to the filter’s $\mathbf{R}_z$; both aiding streams are withheld for $t \in [25, 35]$ s. Seed: 20240517.

Baselines, Metrics, and Hardware. No external baseline (e.g., `nav2_amcl` or `hdl_localization`) is run: the synthetic worlds are designed to exercise PRISM-Loc’s own cores, and a comparative accuracy study on public data is left to future work (Section 5). We report position error (Euclidean, m) and absolute yaw error against synthetic ground truth, BBS success rate (position error < 0.5 m and yaw error $< 10^\circ$), acceptance and false-accept counts at the threshold ρ of Equation (11), windowed RMSE and 3σ consistency for the ESKF, and BBS wall-clock latency. All accuracy numbers are deterministic; latency is machine-dependent and was measured single-threaded on the machine that produced the committed CSV.

4.2. Main Results

MCL Tracking and KLD Adaptation. Figure 2 and Table 2 show that after convergence ($t \geq 10$ s, across all three seeds) the `laser2d` filter holds a mean position error of 0.105 m (max 0.29 m) and a mean yaw error of 0.032 rad, i.e. 1.8° (max 0.18 rad), with little spread across seeds (0.105 ± 0.007 m over per-seed means). The KLD bound (Equation (5)) sizes the set to the posterior: it starts at the 2000-particle ceiling under the broad prior, collapses to the 500-particle floor after the first post-convergence update, and holds there for the rest of the loop, a $4\times$ reduction in per-update likelihood evaluations once the posterior is concentrated.

Global Relocalization. Of the 100 kidnapped-robot queries, 97 succeed (Table 3); over all queries the recovered pose has median position error 0.037 m (95th percentile 0.072 m) and median yaw error 0.48° (95th percentile 1.74°), consistent with the 0.1 m grid and $\approx 1^\circ$ rotation lattice. Figure 3 shows that the score fraction of Equation (11) is an informative confidence signal: the three failures

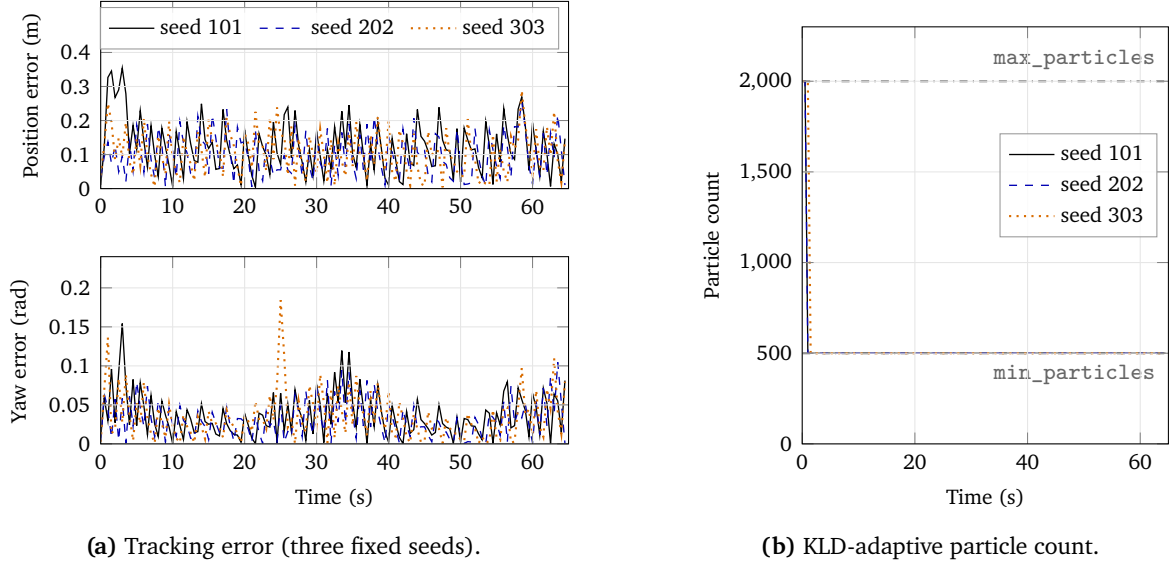


Figure 2: laser2d closed-loop tracking on synthetic data. A simulated robot drives a closed oval loop (semi-axes 3.6 m and 2.6 m) for 65 s at 10 Hz in a 12 m×10 m grid with two symmetry-breaking pillars; the real ParticleFilter with Laser2DLikelihoodField and KLD resampling is updated exactly as in Algorithm 1. (a) Starting from a broad prior, the filter converges within a few updates; across the three seeds the steady-state ($t \geq 10$ s) mean position error is 0.105 m (max 0.29 m) and the mean yaw error 0.032 rad (max 0.18 rad). (b) The particle set starts at the 2000-particle ceiling and collapses to the 500-particle floor after the first post-convergence update, where it stays.

score in 0.21–0.27 with more than 3 m and roughly 90° – 180° error, all below $\rho = 0.40$, so the default threshold yields *zero false accepts* while accepting 78 matches, all correct, and conservatively rejecting 19 correct fixes. The median score fraction is 0.632 and successes span 0.22–0.97, so three low-scoring successes overlap the failure band. Median full-map query latency is 37.65 ms.

ESKF Fusion. With full aiding the windowed position RMSE is 0.016 m (Table 3). During the 10 s blackout the filter dead-reckons on the IMU alone and the RMSE rises to 0.125 m, with a peak error of 0.256 m at $t = 34.5$ s at the end of the blackout; the first aiding cycle brings the error back to 0.015 m by $t = 35$ s, and the post-dropout RMSE is 0.009 m (Figure 4). The filter’s own 3σ bound inflates from roughly 0.03–0.04 m under full aiding to 0.663 m and stays above the true error at every logged sample, i.e. the reported covariance is not overconfident on this single seeded run. This is a one-sided envelope check, not a Monte Carlo NEES consistency test, which remains future work.

4.3. Ablation Studies

The committed runs also isolate the effect of three mechanisms. All numbers below are recomputed from the same CSVs; no additional experiment was run for this section.

Acceptance Threshold. Table 4 sweeps ρ over the recorded score fractions. Without the gate ($\rho \leq 0.20$) all three failures would re-seed the filter at a wrong pose; any $\rho \in [0.30, 0.60]$ removes every false accept, and the shipped $\rho = 0.40$ trades 19 rejected correct fixes for a margin of 0.13 above the highest-scoring failure. Because a rejected match merely waits for the next scan, whereas a false accept silently corrupts the posterior, we keep the conservative default; $\rho = 0.30$ would have accepted 16 more correct fixes on this data.

Clutter. Table 5 shows how clutter affects the matcher. Below a clutter probability of 0.533 every

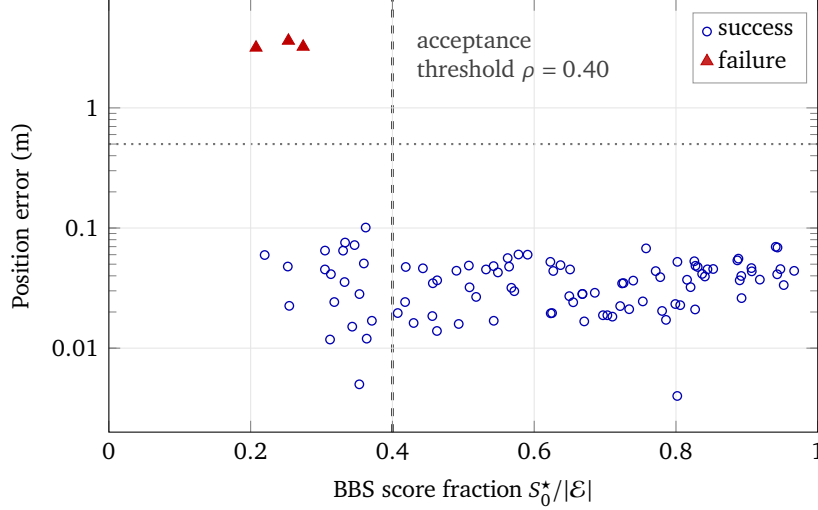


Figure 3: Branch-and-bound relocalization of a kidnapped robot on synthetic data. Each marker is one of $N = 100$ queries in an asymmetric $6\text{ m} \times 6\text{ m}$ room: position error versus the score fraction of Equation (11). Each 180-beam scan is corrupted with a per-query clutter probability drawn uniformly in $[0, 0.8]$. Success means position error $< 0.5\text{ m}$ (dotted) and yaw error $< 10^\circ$. All three failures score below the default threshold (dashed), so every accepted match is correct (zero false accepts); 19 correct fixes are conservatively rejected.

query succeeds and is accepted; in the top third the pose is still recovered in 30 of 33 cases with an unchanged median error, but the median score fraction falls to 0.353, so only 11 are accepted. Clutter therefore degrades the matcher’s *confidence* much faster than its *accuracy*, and the gate absorbs the difference; all three failures occur in this top third.

KLD Adaptation and Aiding. Replacing KLD adaptation with a fixed set at the 2000-particle ceiling would cost four times the steady-state likelihood evaluations of the 500-particle floor; averaged over each whole run, including the initial transient, the adaptive set holds 526.9 particles, $3.8\times$ fewer than the ceiling, with the accuracy of Table 2. The dropout window acts as an ablation of both aiding streams: removing NDT and GNSS updates raises position RMSE from 0.016 m to 0.125 m over 10 s, and restoring them recovers 0.009 m, so the growth is the bounded drift of IMU-only integration from which the filter recovers within one aiding cycle.

5. Discussion and Limitations

What the Factoring Buys. Because the MCL core is written against one `MeasurementModel` interface, the 2D and 3D backends share the motion model, weighting, KLD resampling, estimate, and every test of those components; adding a backend means adding one likelihood function. Because the cores are middleware-free, every equation of Section 3 is exercised by deterministic tests and by the closed-loop drivers of Section 4 without a ROS installation, which is also what makes the evaluation reproducible bit for bit from the repository.

Cost Analysis. The per-update cost of the MCL backends is linear in the particle count and in the number of scored beams or points (Equations (7) and (8)); KLD adaptation keeps the former at the 500-particle floor once the posterior is concentrated (Figure 2). The branch-and-bound matcher’s cost is data-dependent through pruning: on the synthetic room a full-map query took a median of 37.65 ms (min 8.77 ms, 95th percentile 106.07 ms) on the machine that produced the committed data, and it runs only when relocalization is requested, not every scan. The ESKF step is a fixed

Table 3: Relocalization and fusion results on synthetic data. Left: branch-and-bound over $N = 100$ kidnapped-robot queries (data/bbs_relocalization.csv); percentiles use linear interpolation; latency is single-threaded wall-clock on the machine that produced the committed CSV. Right: ESKF position error by window relative to the 10 s aiding dropout (data/eskf_track.csv, logged at 2 Hz).

BBS relocalization	
Success (pos. < 0.5 m, yaw $< 10^\circ$)	97 / 100
Median / p95 position error	0.037 / 0.072 m
Median / p95 yaw error	0.48° / 1.74°
Accepted at $\rho = 0.40$ / false accepts	78 / 0
Median score fraction	0.632
Median latency (min, p95)	37.65 ms (8.77, 106.07)
ESKF fusion	
RMSE before dropout ($1 \leq t < 25$ s)	0.016 m
RMSE during dropout ($25 \leq t < 35$ s)	0.125 m
RMSE after dropout ($t \geq 35$ s)	0.009 m
Peak error (at $t = 34.5$ s)	0.256 m
Peak 3σ bound (at $t = 34.5$ s)	0.663 m
Samples with error $> 3\sigma$	0 / 121

Table 4: Ablation of the BBS acceptance threshold ρ (Equation (11)), recomputed post hoc from the score fractions in data/bbs_relocalization.csv; no search is re-run. “Rejected correct” counts successful matches that the threshold would discard. The shipped default is $\rho = 0.40$.

ρ	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.60
Accepted	100	98	94	84	78	73	67	54
False accepts	3	2	0	0	0	0	0	0
Rejected correct	0	1	3	13	19	24	30	43

15×15 covariance propagation per IMU sample.

Limitations. PRISM-Loc is an architecture paper, and its scope is deliberately bounded. (i) *Planar state in the MCL backends.* Both particle-filter backends estimate (x, y, θ) ; ndt3d evaluates a full 3D Gaussian-voxel map from that planar pose at a fixed base height, which suits wheeled robots on locally flat floors but does not track roll, pitch, or vertical motion. Full $SE(3)$ state exists only in fusion3d. (ii) *2D-only, window-bounded global relocalization.* Algorithm 2 operates on a 2D occupancy pyramid; ndt3d and fusion3d rely on an operator-supplied or GNSS-derived initial pose and local tracking. The search covers the fixed window $\mathbf{c} + \text{res} \cdot \mathcal{O}$ around the map center, so on maps wider than about $2W$ per axis a kidnapped robot outside the window cannot be recovered, and the matcher still returns the best in-window pose; candidates are not restricted to free or on-map cells, and the score fraction of Equation (11) is the only guard against accepting such a pose. The evaluation uses a single 6 m room that the window covers, so neither out-of-window false accepts nor large-map latency has been measured. (iii) *No time-offset or extrinsic estimation.* The ESKF treats the LiDAR/GNSS-to-IMU time offset and the LiDAR-IMU extrinsic as known inputs. (iv) *Loosely-coupled LiDAR update.* In fusion3d, PCL NDT produces a 6-DoF pose measurement that the ESKF fuses as a loosely-coupled update, which keeps the filter core independent of the registration engine but forgoes the accuracy of tightly-coupled formulations such as FAST-LIO (Xu and Zhang, 2021); the implementation also omits the reset Jacobian of Solà (2017). (v) *Single static map.* All backends assume one prior map of a static world, loaded once, with no online map update or change detection. (vi) *No field-accuracy evaluation.* The evaluation of Section 4 is deterministic unit and integration testing plus a seeded

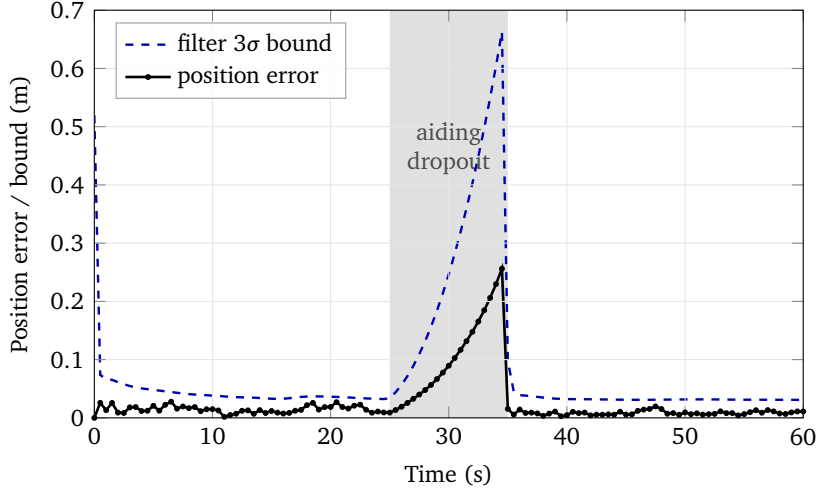


Figure 4: ESKF fusion under a 10 s aiding dropout on synthetic data. A 200 Hz IMU is fused with 10 Hz NDT-like 6-DoF pose and 1 Hz GNSS-like position updates over a 60 s 3D figure-eight; both aiding streams are withheld for $t \in [25, 35]$ s (shaded). The position error grows to its peak of 0.256 m on dead reckoning alone while the filter’s own 3σ bound inflates to 0.663 m, and both collapse within one aiding cycle once updates resume. The bound stays above the true error at every logged sample.

Table 5: Ablation by clutter level. The 100 BBS queries of Figure 3 split into thirds of the clutter range $[0, 0.8]$ by the per-query `clutter_frac` column; success as in Table 3, acceptance at $\rho = 0.40$.

Clutter probability	Queries	Success	Accepted	Median pos. err. (m)	Median score fraction
$[0, 0.267)$	37	37	37	0.040	0.831
$[0.267, 0.533)$	30	30	30	0.032	0.624
$[0.533, 0.8]$	33	30	11	0.037	0.353

synthetic study whose worlds and sensor-noise models match the filters’ assumptions by construction, and it runs no external baseline. It establishes that each core behaves as specified in closed loop, not how accurate PRISM-Loc is on a physical robot. We have not measured trajectory error on physical or public datasets; the planned evaluation will report absolute and relative trajectory error (ATE/RPE) following the TUM RGB-D conventions (Sturm et al., 2012) on the per-backend map $\rightarrow$ odom estimates. Until then we make no accuracy claim.

Future Work. Extending the MCL core to a 6-DoF particle state or coupling it to an external attitude source, a 3D branch-and-bound or place-recognition front end for the 3D backends, joint time-offset and extrinsic estimation in the ESKF, and the public-dataset evaluation above are the next steps on the roadmap.

6. Conclusion

We presented PRISM-Loc, a LiDAR localization stack whose contribution is architectural rather than algorithmic: three known estimator families, likelihood-field MCL, NDT-MCL, and an error-state Kalman filter fusing LiDAR, IMU, and RTK-GNSS, sit behind a single ROS 2 topic and TF contract, so that a 2D LiDAR, a 3D LiDAR, or a sensor-fused configuration is a deployment choice rather than a separate software stack. Each estimator is a pure C++ core with no dependence on the middleware runtime, the two MCL backends share one predict–weight–resample core that differs only

in its measurement model, and `laser2d` relocalizes deterministically by branch-and-bound within a configurable window around the map center. Seeded synthetic experiments show that the assembled pipelines behave as intended in closed loop; a field-accuracy evaluation remains future work. PRISM-Loc is released under the Apache-2.0 license at https://github.com/kjungmo/prism_loc.

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A. Reproduction

Every number and figure of Section 4 regenerates from the repository alone. The drivers and the core test suites need only a C++17 compiler and the Eigen 3 headers (Debian/Ubuntu packages `g++`, `libeigen3-dev`); the unit tests additionally use GoogleTest (`libgtest-dev`). ROS 2 Humble is required only for the two node packages, which continuous integration builds in a `ros:humble` container (`.github/workflows/ci.yml`). From the repository root:

```
CORE="-I prism_loc_core/include -I prism_loc_core/test -I /usr/include/eigen3"
g++ -O2 -std=c++17 $CORE prism_loc_core/src/*.cpp \
    experiments/mcl_tracking.cpp -o /tmp/exp_mcl && /tmp/exp_mcl
g++ -O2 -std=c++17 $CORE prism_loc_core/src/*.cpp \
    experiments/bbs_relocalization.cpp -o /tmp/exp_bbs && /tmp/exp_bbs
g++ -O2 -std=c++17 -I prism_loc_fusion/include -I /usr/include/eigen3 \
    prism_loc_fusion/src/*.cpp experiments/eskf_fusion.cpp -o /tmp/exp_eskf
/tmp/exp_eskf > docs/paper/data/eskf_track.csv
python3 docs/paper/check_numbers.py # recompute every quoted statistic
```

All randomness derives from fixed compile-time seeds (Table 6), so every CSV column is bit-for-bit reproducible except the wall-clock `time_ms` latency column of the BBS experiment. We re-ran all three drivers from the sources of this commit while preparing this version: `mcl_tracking.csv`, `mcl_particles.csv`, and `eskf_track.csv` were byte-identical to the committed files, and `bbs_relocalization.csv` matched in every column except `time_ms`. On that re-run machine (AMD Ryzen 5 5500GT, single thread) the median BBS query latency was 10.26 ms against the 37.65 ms of the committed CSV, which illustrates why latency is reported as machine-dependent; the re-run CSV is committed under `docs/paper/data/rerun_2026-09-28/`. The guard script `docs/paper/check_numbers.py` recomputes each headline statistic from the committed CSVs and fails if a number quoted in the $\LaTeX$ sources, including this package, no longer matches.

Table 6: Experiment drivers, fixed seeds, and committed artifacts. Drivers live under `experiments/`; CSVs under `docs/paper/data/` (copied to `data/` in this package).

Driver	Seed(s)	Figures / tables	CSV(s)
<code>mcl_tracking.cpp</code>	101, 202, 303	Figure 2, Table 2	<code>mcl_tracking.csv</code> , <code>mcl_particles.csv</code>
<code>bbs_relocalization.cpp</code>	20240607	Figure 3, Tables 3 to 5	<code>bbs_relocalization.csv</code>
<code>eskf_fusion.cpp</code>	20240517	Figure 4, Table 3	<code>eskf_track.csv</code>

B. Parameters

Table 7 lists the defaults used throughout the paper; each is a declared ROS 2 parameter documented one-to-one in `PARAMS.md`, with deployed overrides in `prism_loc/params/laser2d.yaml`, `prism_loc/params/ndt3d.yaml`, and `prism_loc_fusion_ros/params/fusion3d.yaml`.

C. Code Map

Table 8 maps each equation and algorithm to the source file that implements it, and Table 9 collects the notation.

Table 7: Default parameters (from the shipped YAML files; symbols as in Section 3).

Parameter	Symbol	Default	Used in
min_particles / max_particles	$N_{\min} / N_{\max}$	500 / 2000	Equation (5)
resample_threshold	τ	0.5	N_{eff} trigger
kld_err / kld_z	$\epsilon_{\text{kld}} / z_{1-\delta}$	0.05 / 2.33	Equation (5)
kld_bin_xy / kld_bin_yaw	–	0.5 m / 0.17 rad	Equation (5)
update_min_d / update_min_a	$d_{\min} / a_{\min}$	0.2 m / 0.2 rad	Algorithm 1
alpha1-alpha4	$\alpha_{1..4}$	0.2	Equation (3)
max_beams	–	60	Equation (7)
z_hit / z_rand / sigma_hit	$z_{\text{hit}} / z_{\text{rand}} / \sigma_{\text{hit}}$	0.5 / 0.5 / 0.2 m	Equations (7) and (9)
likelihood_max_dist	–	2.0 m	Equation (7)
ndt_resolution / voxel_min_points	–	1.0 m / 5	Equation (8)
max_points / base_height	–	500 / 0.0 m	Equation (8)
bbs_linear_window	W	10.0 m	Algorithm 2
bbs_angular_window / bbs_angular_step	–	π / 0.0175 rad	Algorithm 2
bbs_max_depth / bbs_max_beams	D / –	6 / 120	Equations (9) and (10)
bbs_min_score_fraction	ρ	0.4	Equation (11)
sigma_acc / sigma_gyro	σ_a / σ_g	0.01 / 0.001	Equation (12)
sigma_acc_bias / sigma_gyro_bias	$\sigma_{b_a} / \sigma_{b_g}$	$10^{-4} / 10^{-5}$	Equation (12)
pose_pos_std / pose_rot_std	$\sigma_{\text{pos}} / \sigma_{\text{rot}}$	0.1 m / 0.05 rad	Equation (13)
ndt_max_fitness	–	2.0	NDT gate
gnss_min_status / gnss_max_pos_cov	$s_{\min} / c_{\max}$	0 / 25.0 m ²	GNSS gate
transform_tolerance / seed	–	0.1 s / 42	Equation (2) / RNG

Table 8: Where each equation lives. Paths are relative to the repository root; core/ abbreviates prism_loc_core/src/, fusion/ abbreviates prism_loc_fusion/src/, and node/ and fnode/ abbreviate prism_loc/src/localization_node.cpp and prism_loc_fusion_ros/src/fusion_localization_node.cpp.

Equation / algorithm	Source
Equation (2) (map→odom)	node/, fnode/; tests prism_loc/test/test_tf_math.cpp, prism_loc_fusion_ros/test/test_tf_compose.cpp
Equation (3) (motion model)	core/motion_model.cpp
Equations (4) to (6) (filter)	core/particle_filter.cpp
Algorithm 1 (update order, gates)	node/ (runUpdate, onScan)
Equation (7) (likelihood field)	core/measurement_model.cpp, core/occupancy_grid.cpp
Equation (8) (NDT score)	core/measurement_model.cpp, core/ndt_map.cpp
Equations (9) to (11), Algorithm 2	core/bbs.cpp; re-seeding in node/ (onScan)
Equations (12) and (13) (ESKF)	fusion/eskf.cpp, fusion/so3.cpp, fusion/geodetic.cpp
Equation (14), GNSS and NDT gates	fnode/ (attitudeFromAccel, onGnss, onPoints)
Synthetic drivers	experiments/mcl_tracking.cpp, bbs_relocalization.cpp, eskf_fusion.cpp

Table 9: Notation.

Symbol	Meaning	Symbol	Meaning
$\mathcal{B}$	set of backends	$\mathbf{T}_{a,b}$	pose of frame b in frame a
$\mathbf{x} = (x, y, \theta)$	planar particle pose	$w^{[i]}$	particle weight
$\ell(\mathbf{x})$	measurement log-likelihood	N_{eff}	effective sample size
M_k	KLD sample bound, k bins	$d(\cdot)$	distance to nearest obstacle
$\boldsymbol{\mu}_v, \boldsymbol{\Sigma}_v$	NDT voxel mean, covariance	P_h	BBS pyramid level h
$S_h(k, \mathbf{o})$	BBS node score	$\mathcal{E}$	subsampling scan endpoints
ρ	BBS acceptance fraction	D	BBS pyramid depth
$\mathbf{p}, \mathbf{v}, \mathbf{q}$	ESKF position, velocity, attitude	$\mathbf{b}_a, \mathbf{b}_g$	IMU biases
$\delta \mathbf{x} \in \mathbb{R}^{15}$	ESKF error state	$\mathbf{P}$	error covariance
$\mathbf{F}, \mathbf{Q}$	transition, process noise	$\mathbf{H}, \mathbf{R}_z, \mathbf{K}$	update Jacobian, noise, gain